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环境工程专业培养方案

Curriculum for Undergraduate of Environmental Engineering Major

一、专业介绍

环境工程专业成立于1998年，为4年制学士学位授予专业。专业着眼国际前沿，结合国家重大科技需求，以解决水污染、大气污染、土壤污染和固体废弃物污染等环境问题为目标，研究和开发污染防治新技术、新工艺、新方案等。依托矿业工程、冶金工程等优势特色学科，在冶金二次资源高效利用、高浓度难降解工业废水处理、烧结烟气多污染物协同控制、矿冶环境生态修复等领域开展研究工作，取得多项标志性成果。

专业现有教师22人，包括专任教师20人，实验教师2人。

I. Professional Introduction

The environmental engineering major was established in 1998, offering a four-year Bachelor's degree program. Focusing on international forefronts and addressing national technological demands, the major aims to solve environmental issues such as water pollution, air pollution, soil contamination, and solid waste pollution. It engages in the development of new technologies, processes, and solutions for pollution prevention and control. Leveraging strengths in disciplines like mining engineering and metallurgical engineering, its research efforts are concentrated in areas such as efficient utilization of secondary metallurgical resources, treatment of high-concentration refractory industrial wastewater, synergistic control of pollutants in sintering flue gas, and ecological remediation of mining and metallurgical environments, resulting in several landmark achievements.

This major currently has 22 faculty members, including 20 full-time teachers and 2 laboratory instructors.

二、培养目标

针对国家生态环境保护和矿冶行业绿色发展需求，本专业培养德、智、体、美、劳全面发展的社会主义事业建设者和接班人，具有良好的人文素养、职业道德、社会责任感和可持续发展理念，具备环境工程专业领域的基础理论、基本知识和专业技能，具有一定的国际视野及较好的工程实践、协同创新和终身学习的能力，能够胜任区域和矿冶行业的环境污染控制工程设计及运营、环境监测与评价、环境管理等方面工作的高素质应用型人才。期待毕业生在毕业后5年左右达到以下目标：

1. 具有良好的人文社会科学素养、职业和工程伦理道德及社会责任感，能够在工程实践中综合考虑法律政策、社会伦理、环境保护和可持续发展；
2. 能够综合应用数学、自然科学和专业知

境污染控制的工程设计、技术开发、运营管理、环境监测与评价等工作，并具有一定的创新能力；

3. 具有良好的沟通、协调、组织管理能力和团队精神，能在工程项目团队中作为骨干或者领导有效的发挥作用；

4. 具有一定的国际视野和终身学习的意识，能够持续关注国内外环境保护领域的发展趋势和技术进展，适应社会和行业的发展。

II. Training objectives

In response to the national ecological and environmental protection, as well as the needs of green development in the mining and metallurgical industry, this major trains undergraduate students to be the socialist builders and successors with comprehensive development of moral, intelligence, sports, aesthetics and labor. The students should have good humanistic qualities, professional ethics, social responsibility, and a sustainable development concept, and grasp basic theories, basic knowledge and professional skills of environmental engineering. The students should have a certain international vision, good engineering practice, collaborative innovation and lifelong learning capabilities. The students should be capable of environmental pollution control, design and operation, environmental monitoring and evaluation, environmental management of in some regions or in the mining and metallurgical industry. It is expected that graduates will achieve the following goals in about 5 years after graduation:

1. Having good humanities and social science literacy, professional and engineering ethics and social responsibility, and be able to comprehensively consider laws and policies, social ethics, environmental protection, and sustainable development in engineering practice;

2. Having ability to comprehensively apply mathematics, natural sciences and professional knowledge to analyze and solve complex problems in environmental engineering, and being capable of engineering design, technology development, operation management, environmental monitoring and evaluation of regional and mining and metallurgical industry environmental pollution control, and having certain creativity;

3. Having good ability of communication, coordination, organization and management skills, and teamwork spirit, and being able to play an effective role as a backbone or leader in an engineering project team;

4. Having a certain international vision and a sense of lifelong learning, continuously paying attention to the development trend and technological progress in the field of environmental protection at domestic and foreign, and adapting to the development of society and industry.

三、专业特色

依托矿业工程、冶金工程等优势特色学科，环境工程专业形成了“以区域和矿冶行业污染治理为主线，解决复杂环境工程问题为目标，融合现代技术为手段，培养学生工程实践能力和创新创业能力”的本科生培养特色，成为我国中部具有行业特色的环境工程高层次人才培养和科技创新基地。

III. Professional characteristics

Based on the advantages of disciplines such as mining engineering and metallurgical engineering, the environmental engineering major has developed a distinctive undergraduate education model, i.e. "taking regional and mining-metallurgical industry pollution control as the main thread, aiming to solve complex environmental engineering problems, integrating modern technology methods, and fostering students' engineering practical ability as well as the innovation and entrepreneurship capabilities." This major has become a high-level talent cultivation and scientific innovation base in central China with industry characteristics in environmental engineering.

四、毕业要求

1. 工程知识：能够综合运用数学、自然科学、工程基础和环境工程专业知识用于解决环境污染防治工程的设计、运行和管理等复杂环境工程问题。
2. 问题分析：能够应用数学、自然科学和环境工程科学的基本原理，识别和表达区域和矿冶行业的环境工程复杂问题，并结合文献资料研究，分析过程的影响因素，以获得有效结论。
3. 设计/开发解决方案：能够根据区域和矿冶行业环境污染防治要求，对污染防治中的复杂工程问题提出合理的解决方案，设计满足相应的工艺流程，并在设计环节中体现创新意识，考虑社会、健康、安全、法律、文化以及环境等因素。
4. 研究：能够基于环境科学原理，采用科学方法对区域和矿冶行业复杂环境工程问题进行研究，包括设计实验、分析与解释数据，并通过信息综合得到合理有效的结论。
5. 使用现代工具：针对环境工程特别是矿冶行业领域的复杂工程问题，能够开发、选择与使用恰当的现代分析仪器、信息技术工具、资源和软件对工程问题进行模拟和预测，并理解其局限性。
6. 工程与社会：能够基于环境工程相关背景知识进行合理分析，评价环境专业工程实践和复杂环境工程问题解决方案对社会、健康、安全、法律以及文化的影响，并理解应承担的责任。
7. 环境和可持续发展：能够理解并正确评价针对复杂环境工程问题的工程实践对环境、社会可持续发展的影响，并能够对方案进行优化，使其满足可持续发展的要求。
8. 职业规范：具有良好的人文社会科学素养、高度的社会责任感和保护环境的使命感，能够在环境工程实践中理解并恪守工程职业道德和规范，履行环境保护的社会责任。
9. 个人和团队：具备团队协作的精神，能够在多学科背景下的团队中承担个体、团队成员以及负责人的角色。
10. 沟通：能够就复杂的环境工程问题与业界同行及社会公众进行有效沟通和交流，包括撰写报告和设计说明书、陈述发言、清晰表达或回应指令。并具备一定的国际视野，能够在跨文化背景下进行沟通和交流。
11. 项目管理：理解并掌握环境工程管理原理与经济决策方法，并能在多学科环境中应用，具有较强的组织管理能力。
12. 终身学习：具有自主学习和终身学习的意识，有不断学习新知识，拓展新技能，不断增强自身专业水平和适应发展的能力。

IV. Requirements

1. Knowledge of engineering: Being able to comprehensively use mathematics, natural sciences, engineering foundations, and environmental engineering expertise to solve complex issues in environmental engineering such as the design, operation, and management in the prevention and control projects of environmental pollution.
2. Analysis of problem: Being able to apply the basic principles of mathematics, natural sciences and environmental engineering sciences to identify and express complex environmental engineering problems in the region and the mining and metallurgical industry, combined with literature research, and analyze the influencing factors of the process to obtain effective conclusions.
3. Solutions of design/development: According to the prevention and control requirements of environmental pollution in regions or mining and metallurgical industry, developing reasonable solutions to complex engineering issues in pollution prevention and control, meeting the corresponding process flow in the design, and considering the sense of innovation, social, health, safety, legal, cultural, and environmental factors in the design.
4. Research: Based on the principles of environmental science, using scientific methods to study complex

issues of environmental engineering in the region and the mining and metallurgical industry, including designing experiments, analyzing and interpreting data, and obtaining reasonable and effective conclusions through information synthesis.

5. Use of modern tools: Being able to develop, select and use appropriate modern analytical instruments, information technology tools, resources and software to simulate and predict engineering problems, and understand their limitations, during the solving of complex issues of environmental engineering, especially in the mining and metallurgical industry.

6. Engineering and society: Based on the background knowledge of environmental engineering, being able to reasonably analyze and evaluate the effects of solutions to environmental engineering practices and complex issues in environmental engineering on society, health, safety, law, and culture, and figure out the undertaken responsibilities.

7. Environment and sustainable development: Being able to understand and correctly evaluate the impact of engineering practice for complex issues in environmental engineering on the environment and sustainable development of society, and being capable of optimizing the program in order to meet the requirements of sustainable development.

8. Professional norms: Having good humanities and social sciences literacy, a high sense of social responsibility and a sense of mission to protect the environment, being able to understand and obey engineering professional ethics and norms in environmental engineering practice, and fulfill social responsibility for environmental protection.

9. Individuals and teams: Have good spirit of teamwork, and being able to play the roles of individuals, team members, or leaders in a multidisciplinary team.

10. Communication: Being able to effectively communicate with peers and the public on complex issues in environmental engineering, being able to write reports and design scripts, make presentations, clearly express and respond to instructions. Having a certain international vision and being able to communicate under cross-cultural background.

11. Project management: Understanding and mastering the management principles of environmental engineering and economic decision-making methods, which can be applied in a multi-disciplinary environment, and having strong organizational management capabilities.

12. Lifelong learning: Having the consciousness of independent learning and lifelong learning, having the ability to continuously learn new knowledge, expand new skills, improve professional level and adapt to development ability.

附：培养目标实现矩阵

	培养目标 1	培养目标 2	培养目标 3	培养目标 4
毕业要求 1		√		
毕业要求 2		√		
毕业要求 3		√		
毕业要求 4		√		
毕业要求 5		√		√

	培养目标 1	培养目标 2	培养目标 3	培养目标 4
毕业要求 6	√			
毕业要求 7	√			
毕业要求 8	√		√	
毕业要求 9			√	
毕业要求 10			√	√
毕业要求 11		√	√	
毕业要求 12				√

五、专业主干课程

环境工程导论、环境工程原理、环境化学、环境工程微生物学、水污染控制工程、大气污染控制工程、固体废物处理与处置、物理性污染控制、环境监测、环境影响评价、环境热点案例讨论、工程流体力学。

V. Core courses

Introduction to Environmental Engineering, Principles of Environmental Engineering, Environmental Chemistry, Environmental Engineering Microbiology, Water Pollution Control Engineering, Air Pollution Control Engineering, Solid Waste Treatment and Disposal, Physical Pollution Control, Environmental Monitoring, Environmental Assessment, Discussion of Environmental Case, Engineering Fluid Mechanics.

六、基本学制：四年

VI. Recommended length of the program: 4 years

七、授予学位：工学学士

VII. Degree: Bachelor of Engineering

学生修满所规定的最低毕业学分，符合武汉科技大学授予学士学位规定，授予工学学士学位。

八、毕业学分要求：160学分

课程类型		学分要求	课程类型		学分要求
1、公共课程平台		43	3、专业课程模块		47.5
公共基础课程		28	专业必修课程		38.5
通识教育课程	必修	9	专业选修课程		9
	选修	6	4、实践教学模块		20.5
2、学科基础平台		43	专业实践课程	必修	20.5
专业学科基础课程	必修	41			
	选修	2	5、素质拓展模块		6

*通识教育选修课6学分包括：理工类、医学类、人文社科类、经济管理类中选择2学分（学生在本专业所属类别外的3个类中选择2个学分）；思想政治及新时代素质教育类选择2学分（“四史”1个学分，国家安全教育1个学分）；美育教育类选择2学分（美学和艺术史论类、艺术鉴赏和评论类课程至少选修1个学分）。

VIII. Credits required for graduation: 160 credits

Type of courses		Academic credits	Type of courses		Academic credits
1.Common Courses		43	3. Specialized Courses		47.5
Common Basic Courses		28	Required Courses		38.5
General Education Courses	Required Courses	9	Elective Courses		9
	Elective Courses	6	4.Practicum and Internship Courses		20.5
2.General Disciplinary Courses		43	Disciplinary Practical Courses	Required Courses	20.5
Disciplinary Basic Courses	Required Courses	41			
	Elective Courses	2	5.Quality Development Courses		6

九、学分比例

IX. Ratio of Credits

1、必修选修学分比例

The proportion of compulsory elective credits

类别	学分	占总学分比例
必修	143	89.4%
选修	17	10.6%

2、实践教学环节学分比例

The Proportion of credits in practice teaching

类别		学分	占总学分比例
实践教学环节	实验教学学分	24.4	31.813%
	实践教学模块	20.5	
	素质拓展模块	6	

十、辅修

修读本专业辅修课程达到26学分且主修专业达到毕业要求者，颁发辅修专业证书。修读本专业辅修课程达到40学分（含实践教学），并达到辅修学位授予条件的，颁发辅修学位证书。

X. Minor

The minor certificate will be issued to those who have completed 26 credits of minor courses and their own major has met the graduation requirements. The minor degree certificate will be issued to those who have completed 40 credits of minor courses (including practical teaching) and meet the conditions for awarding minor degree.

十一、毕业要求实现矩阵

XI. Graduation Realization Matrix

课程名称	环境工程专业毕业要求											
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
思想道德与法治			√					√				√
中国近现代史纲要								√				
马克思主义基本原理								√				√
毛泽东思想和中国特色社会主义理论体系概论								√				√
习近平新时代中国特色社会主义思想概论								√				√
大学计算机基础					√							
大学英语										√		√
体育									√			
大学生心理健康												√
职业生涯规划与就业指导								√				√
军事课									√			
形势与政策			√					√				
工程制图 B			√		√							
线性代数 B	√											
高等数学 B	√	√										
概率论与数理统计 B	√											
大学物理 B	√											
大学物理实验				√								
无机化学 B	√											
无机化学实验 B				√								
物理化学 B		√										
物理化学实验 B				√								
有机化学 B	√											
有机化学实验 B				√								
分析化学 B	√											
分析化学实验 B				√								
工程力学 B	√	√										

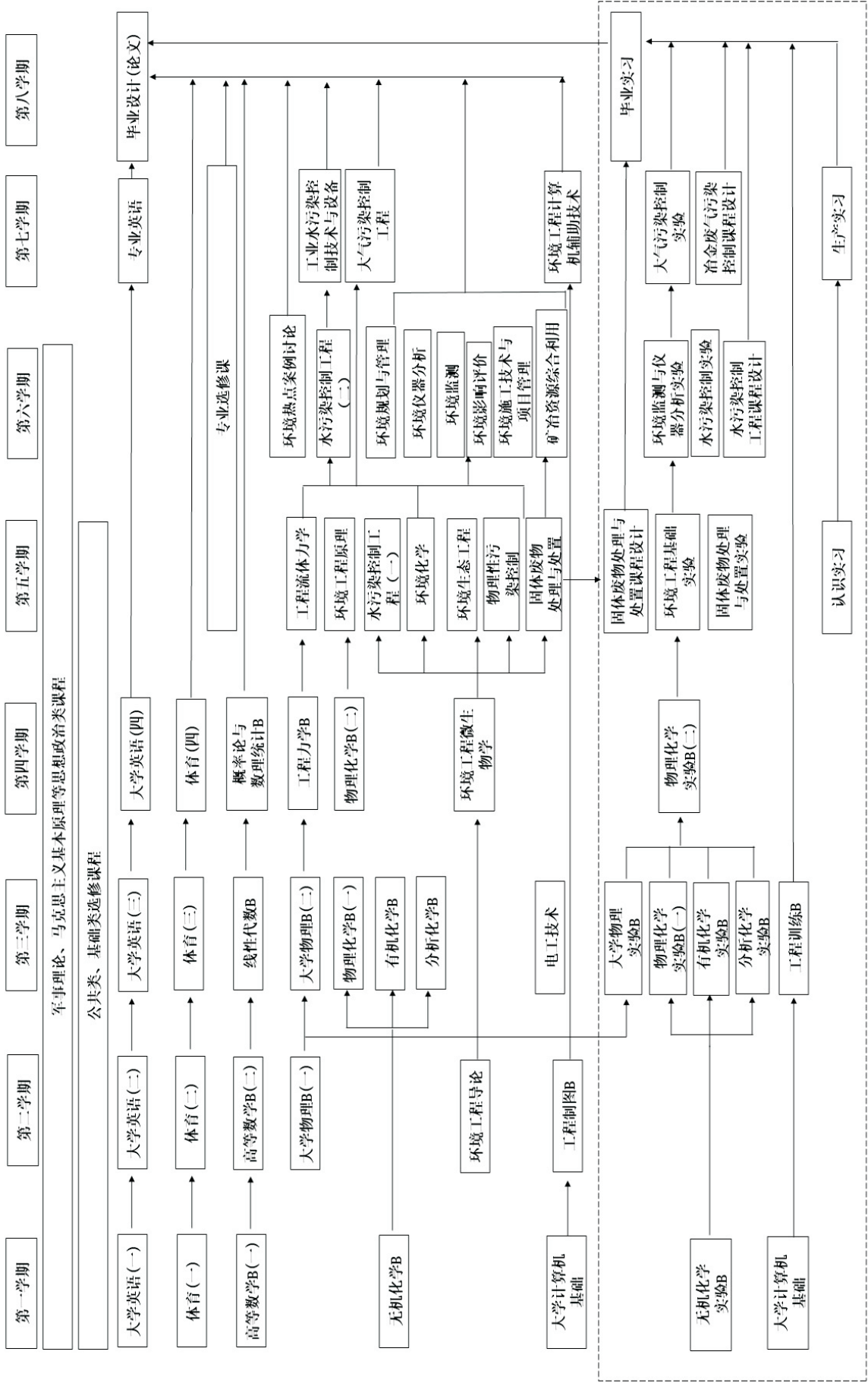
课程名称	环境工程专业毕业要求											
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
电工技术	√											
工程流体力学	√	√										
大气污染控制工程（双语）	√	√	√							√		
环境工程微生物学	√	√										
环境化学	√	√										
水污染控制工程（一）	√	√	√									
水污染控制工程（二）	√	√	√									
物理性污染控制	√	√	√									
环境影响评价					√	√	√					
固体废物处理与处置	√	√	√									
环境监测		√				√						
环境工程原理	√	√	√									
环境工程导论			√					√				√
环境热点案例讨论		√					√		√	√		
环境仪器分析					√							
环境工程基础实验				√					√			
环境监测与仪器分析实验				√					√			
固体废物处理与处置实验				√								
水污染控制实验				√								
大气污染控制实验				√								
环境生态工程			√				√	√				
环境规划与管理						√	√				√	
环境工程计算机辅助技术					√							
矿冶资源综合利用	√		√									
环境施工技术与项目管理									√		√	
工业水污染控制技术与设备			√								√	
冶金废气污染控制课程设计		√	√		√					√		
水污染控制工程课程设计		√	√		√					√		
固体废物处理与处置课程设计		√	√		√					√		
工程训练 B									√			
认识实习						√		√				

课程名称	环境工程专业毕业要求											
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
生产实习						√		√		√		
毕业实习						√	√	√		√		
毕业设计（论文）		√	√	√	√					√	√	√

(四年制本科 2024级 环境工程专业)

十二、课程修读进程表

XII.Course review Process Map



十三、教学环节设置及学分分布表

(四年制本科 2024级 环境工程专业)

XIII. Offered Course and Distribution of Academic Credits

课 程 类 型			课程性质	课 程 编 码	课 程 名 称	学 分	合 计	课 内 学 时			实践学时	学 期	是否辅修/双学位课程	先 修 课 程/备 注
								讲 课	实 验	上 机				
平台	公共基础课程	必修	公共基础课程	5105017	思想道德与法治 Moral Cultivation and Basics of Law	3	48	42	0	0	6	1		
				5103001	中国近现代史纲要 An Outline of Modern and Contemporary History of China	3	48	42	0	0	6	2		
				5102001	马克思主义基本原理 Fundamentals of Marxism	3	48	44	0	0	4	3		
				5101005	毛泽东思想和中国特色社会主义理论体系概论 Theoretical system of socialism with Chinese characteristics	3	48	32	0	0	16	4		
				5101006	习近平新时代中国特色社会主义思想概论 An Introduction to Xi Jinping Thought on Socialism with Chinese Characteristics for a New Era	3	48	32	0	0	16	4		
				1401010	大学英语（一） College English (I)	2	32	32	0	0	0	1		
				1401062	大学英语（二） College English(II)	2	32	32	0	0	0	2		
				1401011	大学英语（三） College English (III)	2	32	32	0	0	0	3		
				1401064	大学英语（四） College English (IV)	1	16	16	0	0	0	4		
				1501882	体育(一) Physical Education (I)	1	26	26	0	0	0	1		
				1501883	体育(二) Physical Education (II)	1	34	34	0	0	0	2		
				1501884	体育(三) Physical Education (III)	1	34	34	0	0	0	3		
				1501885	体育(四) Physical Education (IV)	1	34	34	0	0	0	4		
				5106001	形势与政策 World Affairs and State Policy	2	64	64	0	0	0	1-8		分散进行
				1501024	体质健康测试 Physical Health Test	0	16	0	0	0	16	1-8		分散进行
	通识教育课程	必修		1306009	大学计算机基础 Introduction to Computer Science	2	32	20	0	12	0	1		

教学环节设置及学分分布表

(四年制本科 2024级 环境工程专业)

Offered Course and Distribution of Academic Credits

课 程 类 型	课程 性质	课 程 编 码	课 程 名 称	学 分	合 计	课 内 学 时			实 践 学 时	学 期	是否 辅修/ 双学位 课程	先 修 课 程/备 注
						讲 课	实 验	上 机				
平台	公共课程平台课程	必修	2501016 大学生劳动教育 Community Service	1	32	2	0	0	30	1-4		分散进行
			2502006 大学生心理健康教育 Mental Health Education	2	32	24	0	0	8	1		
			2503001 职业生涯规划与就业指导 Career Plan and Vocational Guidance	1	16	16	0	0	0	2		
			2504005 军事理论 Military Theory	2	36	36	0	0	0	2		
			8001003 大学生创新创业基础 Fundamentals of entrepreneurship	1	16	16	0	0	0	2		
		选修	经济管理类1学分 Economic and Management1 Academic Credit									
			人文社科类1学分 Humanity and Social Sciencel Academic Credit									
			思想政治及新时代素质教育类2学分 Ideological and Political Education2 Academic Credit									
			美育教育类2学分 Art Education2 Academic Credit									
			医学类1学分 Medicine1 Academic Credit									
		必修	0302609 工程制图B Engineering Drawing B	3	48	40	0	8	0	2		
			0701606 工程力学B Engineering Mechanics B	3	48	42	6	0	0	4		
			0702104 线性代数B Linear Algebra B	2	32	32	0	0	0	3		
			0702603 高等数学B(一) Advanced Mathematics B(I)	4	64	64	0	0	0	1		
			0702604 高等数学B(二) Advanced Mathematics B(II)	5	80	80	0	0	0	2		
			0702304 概率论与数理统计B Probability and Mathematical Statistics B	2.5	40	40	0	0	0	4		
			0703605 大学物理B(一) College Physics B(I)	2.5	40	40	0	0	0	2		

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平台	专业学科基础课程	必修	0703606 大学物理B(二) College Physics B(II)	2	32	32	0	0	0	3		
			0703607 大学物理实验B Experiments of College Physics B	1	24	0	24	0	0	3		
			2206665 无机化学B Inorganic Chemistry B	3	48	48	0	0	0	1		
			2253020 无机化学实验B Experiments in Inorganic Chemistry B	1	16	0	16	0	0	1		
			2206675 物理化学B(一) Physical Chemistry B(I)	2	32	32	0	0	0	3		
			2206676 物理化学B(二) Physical Chemistry B (II)	1.5	24	24	0	0	0	4		
			2253054 物理化学实验B(一) Experiments in Physical Chemistry B(I)	1	24	0	24	0	0	3		
			2253026 物理化学实验B(二) Experiments in Physical Chemistry B(II)	1	16	0	16	0	0	4		
			2206679 有机化学B Organic Chemistry B	2.5	40	40	0	0	0	3		
			2253027 有机化学实验B Organic Chemical Experiment B	1	24	0	24	0	0	3		
			2206681 分析化学B Analytical Chemistry B	2	32	32	0	0	0	3		
			2253051 分析化学实验B Analytical Chemical Experiment B	1	24	0	24	0	0	3		
		选修	0401001 电工技术 Electrotechnics	2	32	24	8	0	0	3		
			1306012 数据库技术及应用 Database Technology and Applications	1.5	24	24	0	0	0	3		
			1306013 数据库技术及应用实验 Database Technology and Application Experiment	1	24	0	0	24	0	3		
			1306014 网络技术及应用 Network Technology and Application	1.5	24	24	0	0	0	4		

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平台	学科基础平台	专业基础课程	选修	1306015	网络技术及应用实验 Network Technology and Application Experiment	1	24	0	0	24	0	4		
模块		专业必修课程	必修	0106081	环境工程导论 Introduction to Environmental Engineering	1	16	16	0	0	0	2		辅修课程
				0106125	环境工程原理 Principles of Environmental Engineering	2	32	32	0	0	0	5		辅修课程
				0106093	工程流体力学 Engineering Fluid Mechanics	2	32	32	0	0	0	5		辅修课程
				0106094	环境工程微生物学 Environmental Engineering Microbiology	2	32	32	0	0	0	4		辅修课程
				0106095	环境化学 Environmental Chemistry	2	32	32	0	0	0	5		辅修课程
				0106082	水污染控制工程（一） Water Polution Control Engineeing(I)	2	32	32	0	0	0	5		辅修课程
				0106097	水污染控制工程（二） Water Polution Control Engineeing(II)	3	48	48	0	0	0	6		辅修课程
				0106098	物理性污染控制 Physical Pollution Control	2	32	32	0	0	0	5		辅修课程
				0106126	固体废物处理与处置 Solid Waste Treatment and Disposal	2.5	40	40	0	0	0	5		辅修课程
				0106099	环境影响评价 Environmental Assessment	2	32	32	0	0	0	6		辅修课程
				0106060	环境监测 Environmental Monitoring	2.5	40	40	0	0	0	6		辅修课程
				0106080	环境热点案例讨论 Discussion of Environmental Case	1.5	24	24	0	0	0	6		
				0106070	环境仪器分析 Environmental Instrument Analysis	1.5	24	24	0	0	0	6		辅修课程
				0106100	大气污染控制工程 Air Pollution Control Engineering	2.5	40	40	0	0	0	7		辅修课程
				0106101	环境规划与管理 Environmental Planning and Management	2	32	32	0	0	0	6		辅修课程

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								讲课	实验	上机						
模块		专业必修课程	必修	0106117	环境工程基础实验 Basic Experiment of Environmental Engineering	2	32	0	32	0	0	5				
				0106119	固体废物处理与处置实验 Solid Waste Treatment and Disposal Experiment	1	16	0	16	0	0	5				
				0106128	水污染控制实验 Water Pollution Control Experiment	2	32	0	32	0	0	6				
				0106121	大气污染控制实验 Air Pollution Control Experiment	1	16	0	16	0	0	7				
				0106118	环境监测与仪器分析实验 Environmental Monitoring and Instrumental Analysis Experiment	2	32	0	32	0	0	6				
		专业课程模块		专业选修课程	选修	0106103	环境生态工程 Environmental ecological engineering	1.5	24	24	0	0	0	5		
						0106122	矿冶资源综合利用 Comprehensive Utilization of Mining and Metallurgical Resources	1.5	24	24	0	0	0	6		
						0106105	工业水污染控制技术与设备 Industrial Water Pollution Control Technology and Equipment	1.5	24	24	0	0	0	7		
						0106123	环境施工技术与项目管理 Environmental Civil Construction and Project Management	1.5	24	24	0	0	0	6		
						0106107	环境工程计算机辅助技术 Computer Aided Technology of Environmental Engineering	1.5	24	12	0	12	0	7		
					0106110	环境法 Environment Law	1.5	24	24	0	0	0	7			
					0106111	专业英语 Specialized English	1.5	24	24	0	0	0	7			
					0106131	环境过程与建模方法 Environmental Processes and Modeling Methods	1.5	24	24	0	0	0	7			
					0106113	清洁生产 Clean Production	1.5	24	24	0	0	0	7			
					0106114	大气污染控制设备与设计 Air Pollution Control: Equipment and Design	1.5	24	24	0	0	0	7			

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模块	专业课程模块	专业选修课程	选修	0106115	环保设施运营管理 Operation and Management of Environmental Protection Facilities	1.5	24	24	0	0	0	7		
	实践教学模块	专业实践课程	必修	2504006	军事训练 Military Training	2	112	0	0	0	112	1		
				0106088	认识实习 Introductory Practice Experience	2	2(周)	0	0	0	2(周)	5		
				0106089	生产实习 Production Practice	2	2(周)	0	0	0	2(周)	7		
				0106090	毕业实习 Pre-graduation Internship	2	2(周)	0	0	0	2(周)	8		
				0106091	毕业设计（论文） Undergraduate Project (Thesis)	8	15(周)	0	0	0	15(周)	8		辅修课程
				0106124	冶金废气污染控制课程设计 Design on Metallurgical Waste Gas Pollution Control	1	2(周)	0	0	0	2(周)	7		辅修课程
				0106108	水污染控制工程课程设计 Design on Water Pollution Control	1	2(周)	0	0	0	2(周)	6		辅修课程
				0106109	固体废物处理与处置课程设计 Design on Solid Waste Treatment and Disposal	1	2(周)	0	0	0	2(周)	5		辅修课程
				1701008	工程训练B Engineering Training B	1.5	3(周)	0	0	0	3(周)	3		
模块	素质拓展模块	创新创业教育	必修	创新创业实践3学分 Innovation Practices 3 Academic Credits										
		第二课堂	必修	第二课堂3学分 Second Classroom 3 Academic Credits										

*创新创业教育和第二课堂主要安排在小学期进行